Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 3

Lecture 5

5.1 Two-qubit systems

5.2 Two qubit example – independent Alice and Bob

5.3 General two qubit states: measurement and operations

Lecture 6

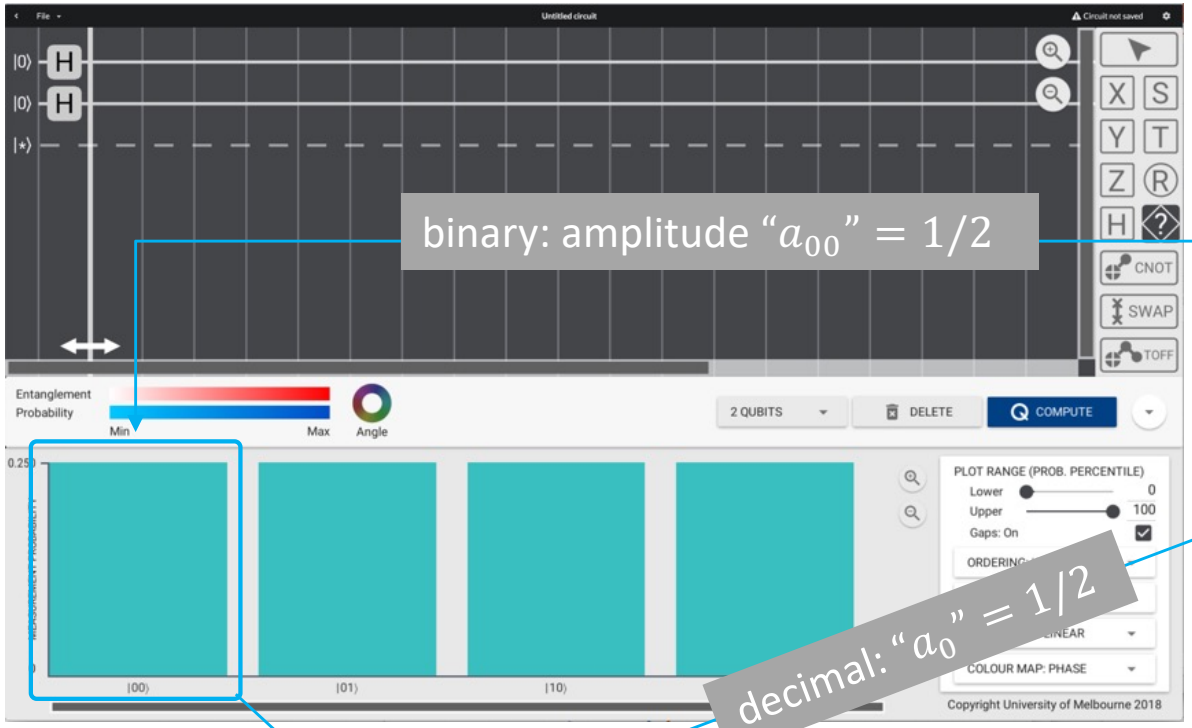
6.1 Two-qubit logic operations

6.2 Entanglement

Practice class 3

Two qubit states and operations

Lecture 5 recap: binary and decimal in the QUI



binary: amplitude "a_{00}" = 1/2

decimal: "a_0" = 1/2

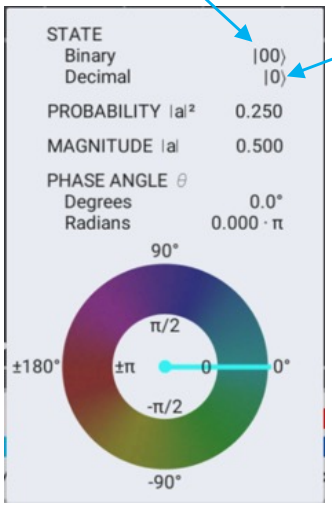
$$|\psi_{\text{alice\&bob}}\rangle = |\psi_{\text{alice}}\rangle \otimes |\psi_{\text{bob}}\rangle$$

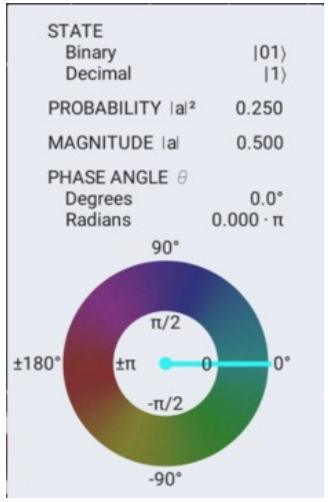
$$= \frac{1}{2}\boxed{|00\rangle} + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}\boxed{|11\rangle}$$

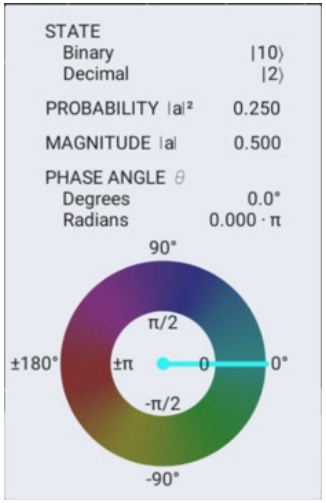
binary amplitude "a_{11}" = 1/2

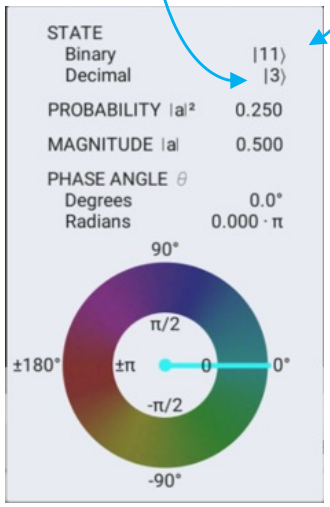
$$= \frac{1}{2}\boxed{|0\rangle} + \frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle + \frac{1}{2}\boxed{|3\rangle}$$

decimal: "a_3" = 1/2









Lecture 5 recap: two-qubit systems

Measurement/collapse on single qubit, renormalisation:

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$

"0" on 1st

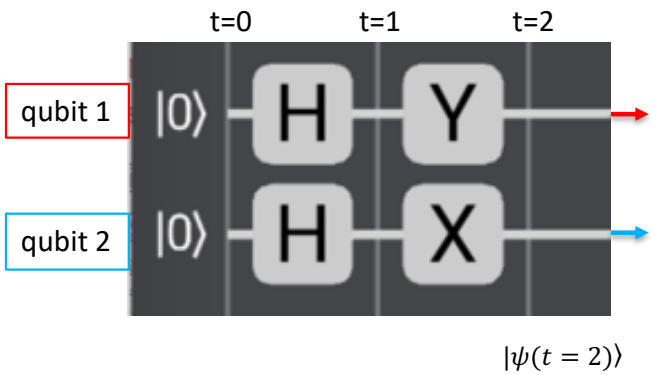
$$|\psi'\rangle = \frac{a\,|00\rangle + b\,|01\rangle}{\sqrt{|a|^2 + |b|^2}}$$

"1" on 1st

$$|\psi'\rangle = \frac{c\,|10\rangle + d\,|11\rangle}{\sqrt{|c|^2 + |d|^2}}$$

& similar for measurements on qubits #2

Single qubit operations on multi-qubit states, e.g. two-qubits):



We write as: $|\psi(t=2)\rangle = Y_1 H_1 X_2 H_2 |00\rangle$ where

H_1 applies a H-gate to qubit #1 in t=0->1, *then*
Y_1 applies a Y-gate to qubit #1 in t=1->2

H_2 applies a H-gate to qubit #2 in t=0->1, *then*
X_2 applies a X-gate to qubit #2 in t=1->2

NB. Order for *different* qubit operators doesn't matter

i.e. we can write $|\psi(t=2)\rangle = X_2 H_2 Y_1 H_1 |00\rangle$

6.1 Two qubit logic operations

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Lecture 6

Recall: Key concepts for quantum computing

Quantum superposition:

Systems can be in indeterminate (multiple) states prior to measurement

Quantum measurement:

Result of any given measurement a-priori unknown, system “collapses” to an outcome

Quantum entanglement:

Systems can be linked such that measurement of one part correlates to that of another part

We’ve looked at superposition and measurement, now for *entanglement*
– while it might seem to be a niche interest for physicists and philosophers,
it is actually a crucial aspect of quantum computing…

Entanglement is generated when the qubits interact – e.g. via two-qubit logic operations

Computing with our quantum register

So far, we considered two qubits each in a superposition, and although we merge the description into a "quantum binary register" you've seen the qubits are still independent

i.e. measurement outcomes for the Alice qubit didn't depend on those of the Bob qubit (& vice-versa)

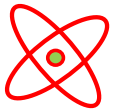
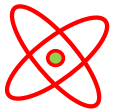
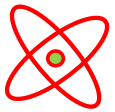
Quantum register state:

$$|\psi_{\text{alice\&bob}}\rangle = \underbrace{\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)}_{|0\rangle \text{ or } |1\rangle} \otimes \underbrace{\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)}_{\text{nothing happens to Bob}} = \underbrace{\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle}_{\text{corresponding quantum register: numbers only, no computing (yet)}}$$

In order to actually *compute* anything with the numbers in our quantum register, we need to *interact* our qubits physically (similar in a sense to classical logic operations).

When we do so, we generate *entanglement* – during the computation the numbers represented in the qubit register become quantum connected…in a sense, they know about each other.

Recall from Lecture 1: Multi qubits – state counting

		Binary combinations
1 qubit:	$\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0\rangle, \lvert 1\rangle$ binary representation of decimals 0 to 1
2 qubits:	 $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 00\rangle, \lvert 01\rangle, \lvert 10\rangle, \lvert 11\rangle$ binary representation of decimals 0 to 3
3 qubits:	 $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 000\rangle, \lvert 001\rangle, \lvert 010\rangle, \lvert 011\rangle$ $\lvert 100\rangle, \lvert 101\rangle, \lvert 110\rangle, \lvert 111\rangle$ binary representation of decimals 0 to 7
4 qubits:	 $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$ $\lvert 0\rangle, \lvert 1\rangle$	$\lvert 0000\rangle, \lvert 0001\rangle, \lvert 0010\rangle, \lvert 0011\rangle$ $\lvert 0100\rangle, \lvert 0101\rangle, \lvert 0110\rangle, \lvert 0111\rangle$ $\lvert 1000\rangle, \lvert 1001\rangle, \lvert 1010\rangle, \lvert 1011\rangle$ $\lvert 1100\rangle, \lvert 1101\rangle, \lvert 1110\rangle, \lvert 1111\rangle$ binary representation of decimals 0 to 15

Multiple qubits and quantum processing

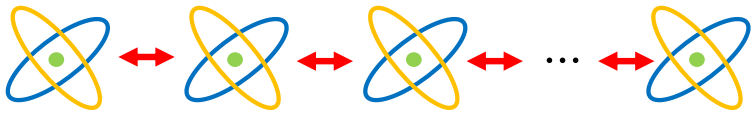
Basic representation of binaries as quantum information:

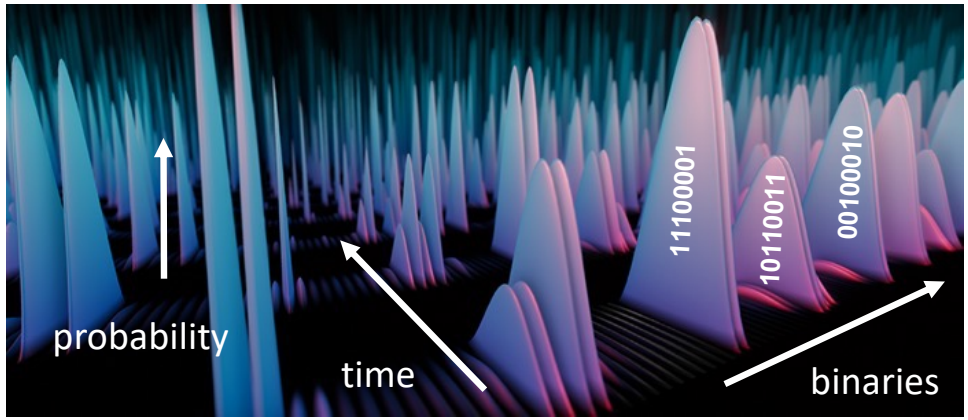
N qubits

Independent quantum superpositions → superposition over N-bit binaries $|000\ldots0\rangle,\ldots,\ |111\ldots1\rangle$ (and there are 2^N of these)

Not very useful…measurement of qubits collapses to one random N-bit string

Quantum computation: qubits interact to create complex superpositions and entangled states



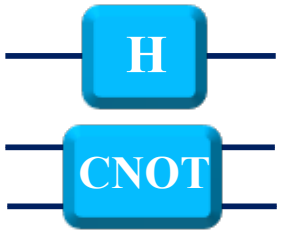


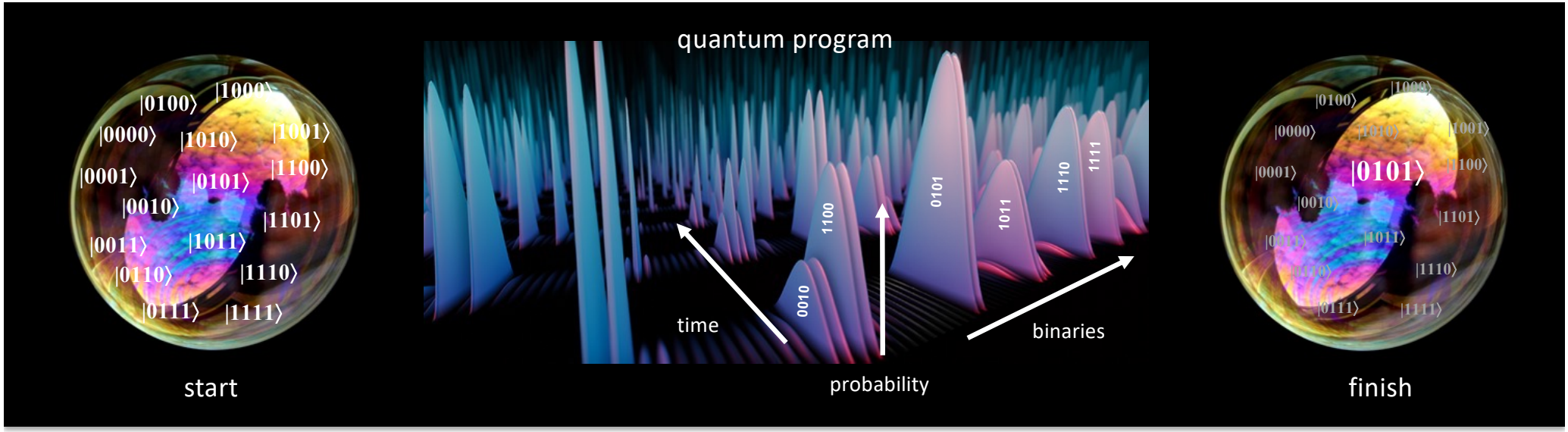
Bubble: Brocken commons.wikimedia.org

Quantum information processing

Digital quantum computing:

- logic gates between qubits perform mathematical operations on binary data
- complex entangled states created → binary data are quantum "linked"
- quantum interference amplifies probability of desired output (answer)





Bubble: Brocken commons.wikimedia.org

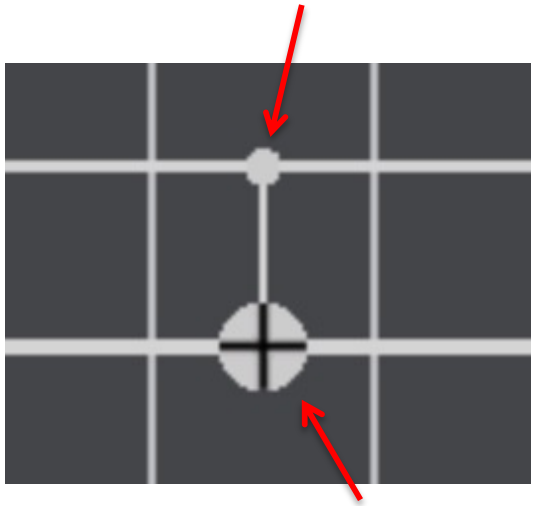
Two qubit logic gates: CNOT

Two qubit gates can be constructed using an interaction between the two systems. Most important is the Controlled-NOT (CNOT) gate.

Symbol for "control"

Control qubit

Target qubit



Symbol for binary addition (bit flip)

How states transform: CNOT truth table

$$|00\rangle \to |00\rangle$$
$$|01\rangle \to |01\rangle$$
$$|10\rangle \to |11\rangle$$
$$|11\rangle \to |10\rangle$$

Rule: The target is flipped iff the <u>control</u> qubit is "1"

$$a\,|00\rangle + b\,|01\rangle + \boxed{c\,|10\rangle + d\,|11\rangle}$$
$$\to a\,|00\rangle + b\,|01\rangle \boxed{+\,d\,|10\rangle + c\,|11\rangle}$$

NB. Here, control qubit is in first position in the kets

CNOT in matrix form

Recall, matrix representation of two qubit states:

$$|\psi\rangle = \underbrace{a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle}_{\text{ket form}} = \begin{bmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{bmatrix} \quad \text{matrix form}$$

CNOT operation: $|\psi\rangle \xrightarrow{CNOT} |\psi'\rangle$

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle \longrightarrow$$

CNOT

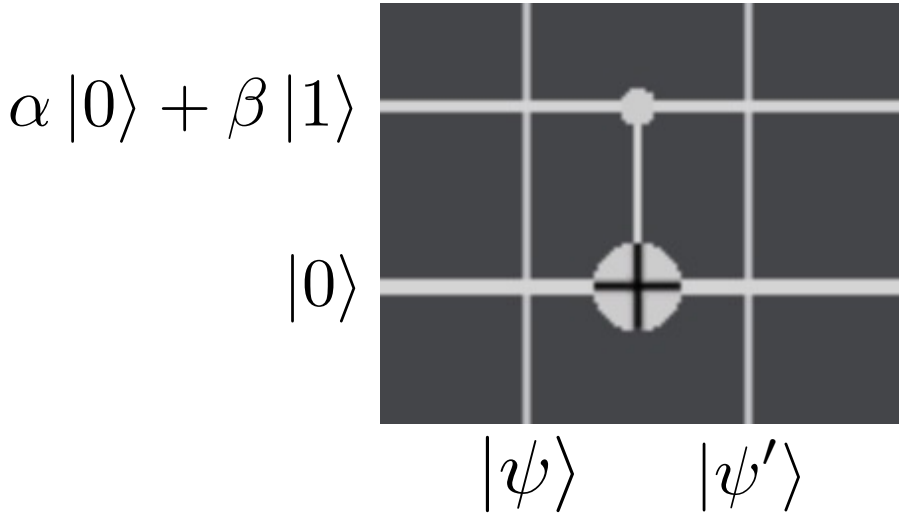
$$\longrightarrow |\psi'\rangle = a_{00}|00\rangle + a_{01}|01\rangle + \underbrace{a_{10}|11\rangle + a_{11}|10\rangle}_{\text{amplitudes flipped}}$$

As a matrix, the CNOT operation is:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$|\psi'\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \overbrace{\begin{bmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{bmatrix}}^{|\psi\rangle} = \begin{bmatrix} a_{00} \\ a_{01} \\ a_{11} \\ a_{10} \end{bmatrix} \text{ amplitudes flipped}$$

Example: CNOT on superposition



Before the CNOT, the state is:

$$|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle = \alpha|00\rangle + \beta|10\rangle$$

After the CNOT, the state is:

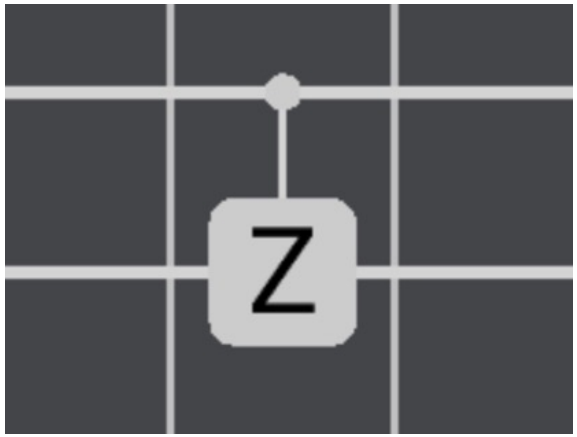
$$|\psi'\rangle = \alpha|00\rangle + \beta|11\rangle$$

Control Phase Gate

Another useful gate is the controlled phase gate (and generalisations using R-gate menu):

Control qubit

Target qubit



How states transform:

$$\begin{aligned} |00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |01\rangle \\ |10\rangle &\rightarrow |10\rangle \\ |11\rangle &\rightarrow -|11\rangle \end{aligned}$$

$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle \boxed{+\,d\,|11\rangle}$$
$$\rightarrow a\,|00\rangle + b\,|01\rangle + c\,|10\rangle \boxed{-\,d\,|11\rangle}$$

Rule: the **phase** of the target flipped iff the control qubit is “1”.

As a matrix:

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Fun fact: CZ matrix is “diagonal” so it doesn’t matter which one you think of as control/target.

SWAP gate

A SWAP operation can be implemented using an interaction between the two qubits – the states of the two qubits are swapped (not the physical qubits).

Qubit 1

Qubit 2

How states transform:

$$|00\rangle \rightarrow |00\rangle$$
$$|01\rangle \rightarrow |10\rangle$$
$$|10\rangle \rightarrow |01\rangle$$
$$|11\rangle \rightarrow |11\rangle$$

$$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$$
$$\rightarrow a\,|00\rangle + c\,|01\rangle + b\,|10\rangle + d\,|11\rangle$$

Rule: the two qubits are swapped.

As a matrix:

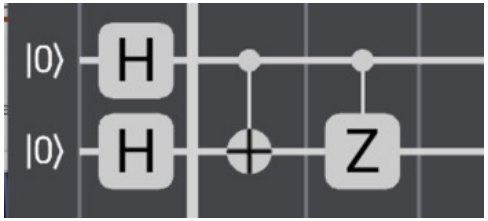
$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

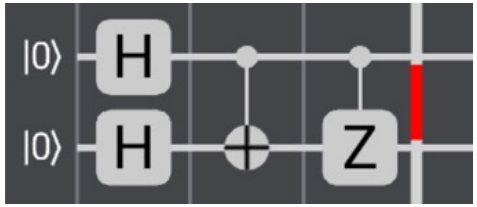
NB. Unlike CNOT, SWAP gates do not generate entanglement (but “sqrt SWAP” does!)

Example: a two-qubit circuit

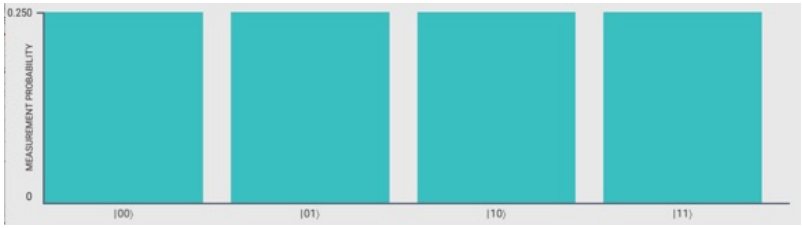
Consider a simple two-qubit circuit:

t=0 t=1 t=2 t=3

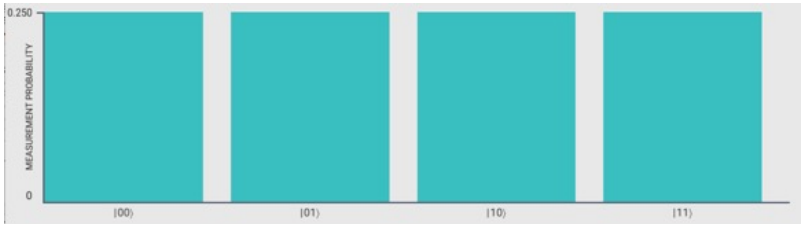




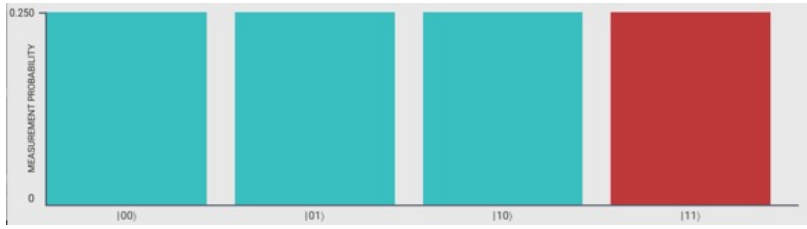
Slider at t=1



Slider at t=2



Slider at t=3



What's gone on here? – the 11 component turned red, as did the slider…

Example: a two-qubit circuit

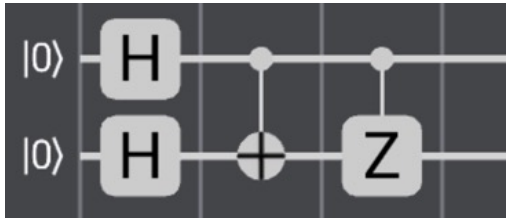
Consider a simple two-qubit circuit:

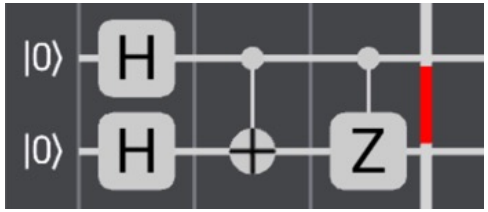
CZ:

$a\,|00\rangle + b\,|01\rangle + c\,|10\rangle \boxed{+\,d\,|11\rangle}$

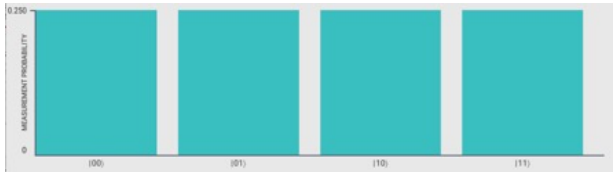
$\rightarrow a\,|00\rangle + b\,|01\rangle + c\,|10\rangle \boxed{-\,d\,|11\rangle}$

t=0 t=1 t=2 t=3

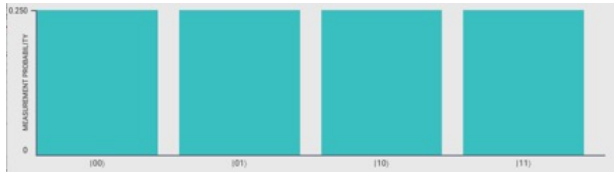




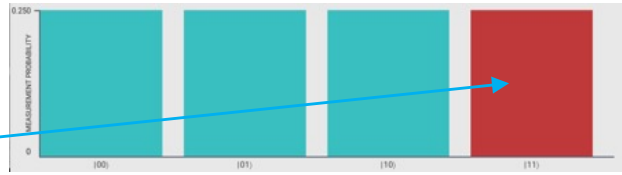
State at t=1



State at t=2



State at t=3



$$|\psi(t=1)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

symmetric -> no change in flip

CNOT:

$$|\psi(t=2)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

CZ:

$$|\psi(t=3)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle \boxed{-}\frac{1}{2}|11\rangle$$

-1 <-> π phase

But, why has the slider turned red? – entanglement!

6.2 Entanglement

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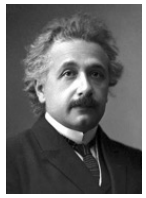
Lecture 6

Entanglement – the real magic of quantum mechanics

So far, we have focused on qubits in quantum superpositions and subsequent measurement.

Strange and wonderful as that is, superposition is actually the least weird aspect of qubits.

Now we get to the stuff that really bothered Einstein and the like – <u>entanglement!</u>

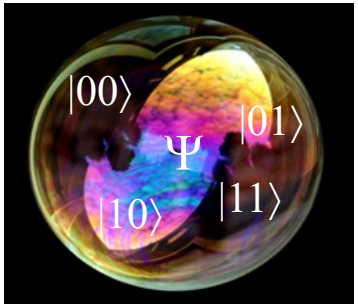


Too "spooky" to be true

Let's look at different types of two-qubit states: $|\psi_{\text{alice\&bob}}\rangle$

Independent superpositions of Alice and Bob

$$\frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

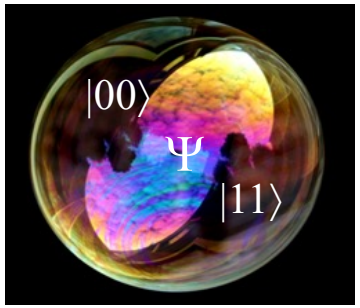


$$|\psi_{\text{alice\&bob}}\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$ "product"

$\rightarrow$ measurement results are independent

An <u>entangled</u> state of Alice and Bob

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$



$|\psi_{\text{alice\&bob}}\rangle$

Looks simpler, but it's far more interesting!

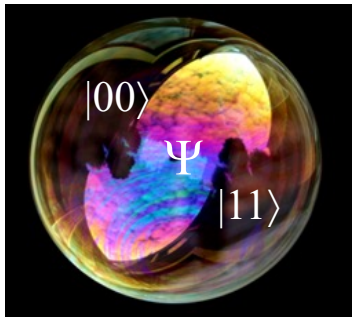
no "product" analogy possible!

$\rightarrow$ entangled, measurement results are dependent!

Entanglement – why it is strange

Let's look at that entangled state in more detail – measurement scenarios:

$|\psi_{\text{alice\&bob}}\rangle$



$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

(NB. this is called a "Bell State" after John Bell who figured out how to really test all this…)

Looks what happens when Alice measures (first qubit)

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

?

50% 50%

Alice measures "0" (50% of the time)

overall state collapses

$\rightarrow |00\rangle$

?

Bob measures "0" 100% of the time

Alice measures "1" (50% of the time)

overall state collapses

$\rightarrow |11\rangle$

?

Bob measures "1" 100% of the time

$a_{00} = 1/\sqrt{2}$

$a_{11} = 1/\sqrt{2}$

Prob[00] = $a_{00}^2 = \frac{1}{2}$

Prob[11] = $a_{11}^2 = \frac{1}{2}$

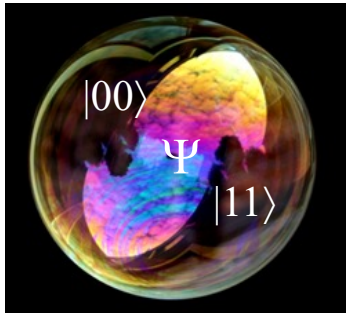
Bob's measurement result is completely determined after Alice's qubit measurement!

i.e. Alice measurement somehow determines Bob's outcome!

Wow.

Aside: why did entanglement worry Einstein?

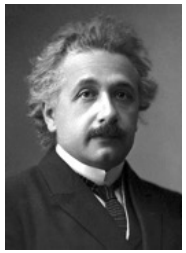
$|\psi_{\text{alice\&bob}}\rangle$



$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Bob's measurement result is completely determined by the result of Alice's qubit measurement!

i.e. Alice measurement determines Bob's outcome!



Einstein (and others) objected to this because it seemed to imply action or communication faster than light…

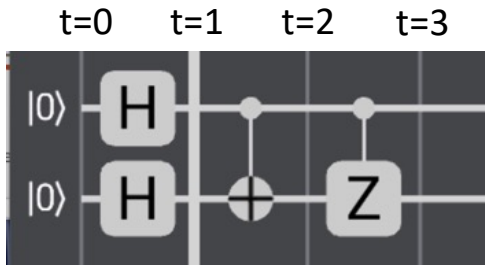
i.e. imagine Alice and Bob qubits are separated to either side of the galaxy…you can see how it appears that Alice sends a signal to Bob in order for Bob's measurement to work out.

Important – even though entanglement worried Einstein, you can prove that no actual communication actually happens between Alice and Bob…it's more like an "influence".

We don't really understand it, but "quantum connections" do occur (Bell showed the way)

More to the point, they are a "resource" that powers quantum computation…

Back to the example circuit

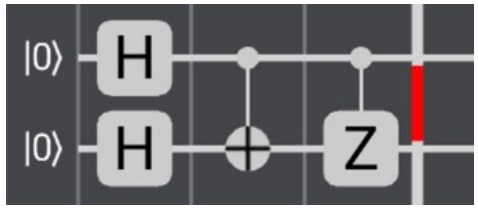


$$|\psi(t=1)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Can write as a “product” -> no entanglement

$$|\psi(t=2)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Can write as a “product” -> no entanglement



$$|\psi(t=3)\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

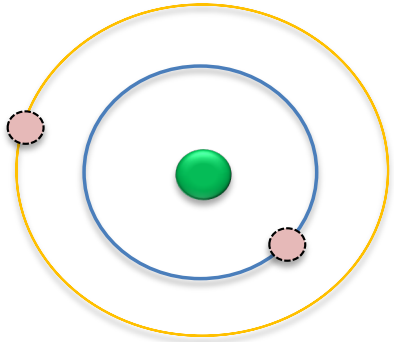
Can’t write as a “product” -> entangled!

Most will say, yes I can I’ll just use i to get the $-\frac{1}{2}|11\rangle$ …but it won’t work…

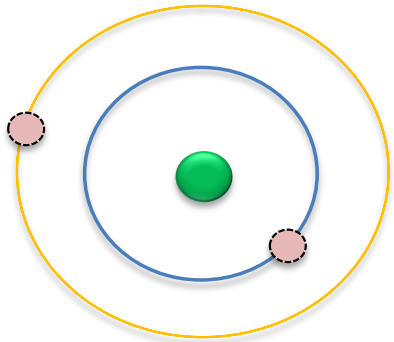
i.e. $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) = \frac{1}{2}|00\rangle + \frac{i}{2}|01\rangle + \frac{i}{2}|10\rangle - \frac{1}{2}|11\rangle$

See, it’s not the same state…

Separable states



$$|\psi\rangle = a|0\rangle + b|1\rangle$$



$$|\phi\rangle = c|0\rangle + d|1\rangle$$

A separable state is one which can be written as

$$|\Phi\rangle = |\psi\rangle \otimes |\phi\rangle$$

All separable states (of two qubits) can be written as:

$$|\psi\rangle = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle$$

Examples of separable states

Consider the state: $$|\psi\rangle = \frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

It is *separable* because: $$|\psi\rangle = |0\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Consider the state: $$|\psi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}$$

It is also *separable* because: $$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Constructing a Bell state

Refers to maximally entangled 2 qubit states named after the physicist John Bell (who figured out how to experimentally explore the reality of entanglement).

Consider the following circuit in the QUI:

Execution:

$$|00\rangle \underset{\text{H}}{\rightarrow} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \underset{\text{CNOT}}{\rightarrow} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Question: Is $\dfrac{|00\rangle + |11\rangle}{\sqrt{2}}$ separable?

Entanglement

Answer: No! We can never find a, b, c, d, i.e.

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}} \neq (a\,|0\rangle + b\,|1\rangle) \otimes (c\,|0\rangle + d\,|1\rangle)$$

A state which is not separable is called an **entangled** state.

Entanglement is a uniquely quantum mechanical property, with no direct classical analogue.

Entanglement Measure

We would like to have a measure of *how much* entanglement a state has. Some states are more entangled than others:

$\|00\rangle$	Not entangled, separable
$\sqrt{0.99}\,\|00\rangle + \sqrt{0.01}\,\|11\rangle$	Entangled, but close to a separable state
$\frac{1}{\sqrt{2}}\,\|00\rangle + \frac{1}{\sqrt{2}}\,\|11\rangle$	Maximally entangled

Entanglement is a type of correlation between two systems, say A and B.

To see how much correlation there is between A and B: We can measure B and ask how many bits of information (as measured by entropy) this can tell us about the state of A?

In the QUI we measure the degree of entanglement using an informatic "entropy" measure: *Entanglement Entropy (EE)*

Entanglement Entropy – a recipe (used on QUI)

Consider an arbitrary normalized two-qubit state:

$$|\psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$$

To calculate the *entanglement entropy* between two qubits, follow these steps…

1. Construct the matrix from the amplitudes: $A = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}$

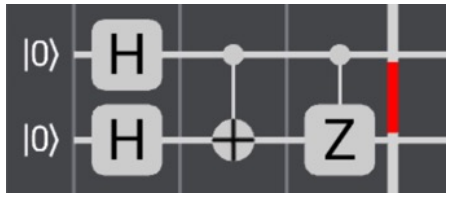
2. Calculate the eigenvalues λ_1 and λ_2 of the matrix product $AA^\dagger$.

3. The entanglement entropy (EE) between the qubits (in number of bits) is given by:

$$H = -\lambda_1 \log_2(\lambda_1) - \lambda_2 \log_2(\lambda_2).$$

NB. Two-qubits: max EE = *1 bit*

QUI does this for you, EE displayed on red-scale on time-slider -> generalises to circuits involving more qubits (EE between qubits above/below the corresponding segment on slider)

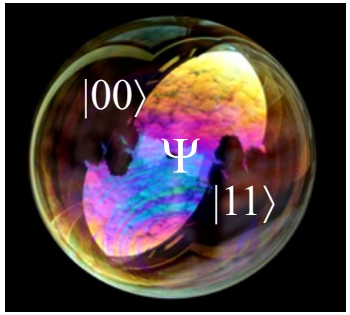


Generating Bell states

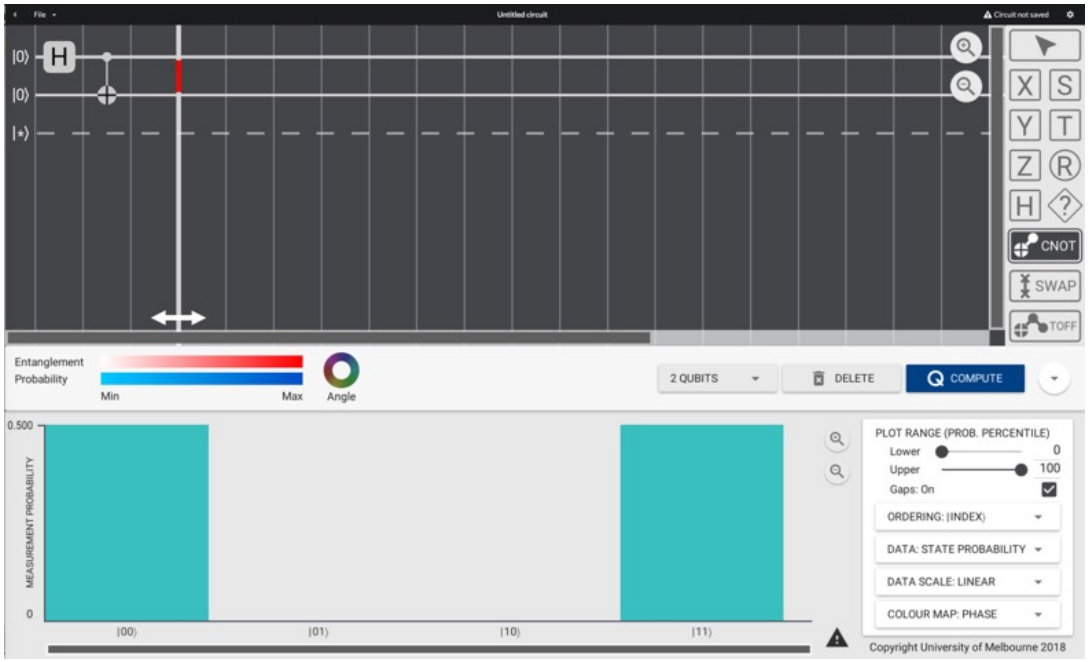
John Bell (1964) figured out a protocol to actually measure whether entanglement was true or not in physical systems (answer is yes to many standard deviations).

The family of simple two-qubit maximally entangled states are named in his honour – *Bell States*

$$|\psi_{\text{alice\&bob}}\rangle$$



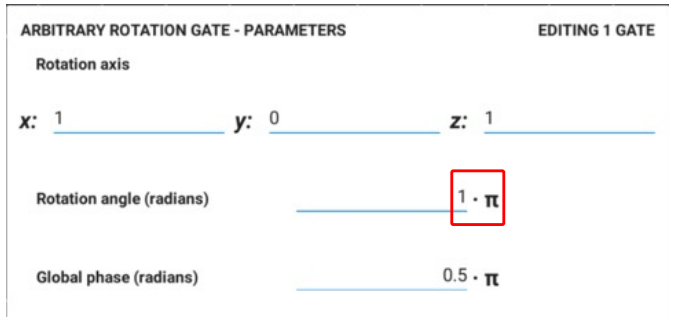
$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$



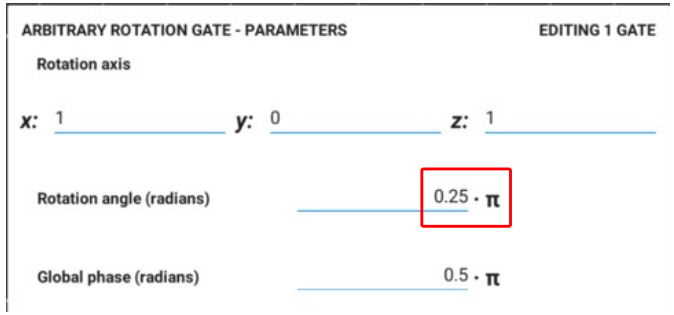
NB. Two-qubits: max EE = *1 bit*

Dialing in the level of entanglement entropy (EE)

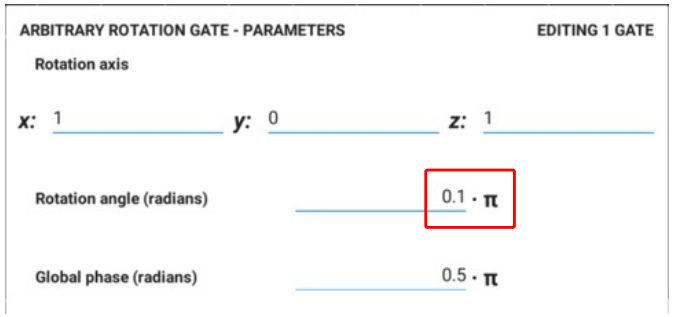
Adjust angle of R-gate rotation (about X + Z axis)



R-gate same as H-gate
-> maximal entanglement



As we reduce
the rotation angle
in R we make the
superposition in
qubit #1 more
unbalanced
before the CNOT



-> slider bar becomes
less red…
(lower entanglement)

Week 3

Lecture 5

5.1 Two-qubit systems

5.2 Two qubit example – independent Alice and Bob

5.3 General two qubit states: measurement and operations

Lecture 6

6.1 Two-qubit logic operations

6.2 Entanglement

Practice class 3

Two qubit states and operations

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8